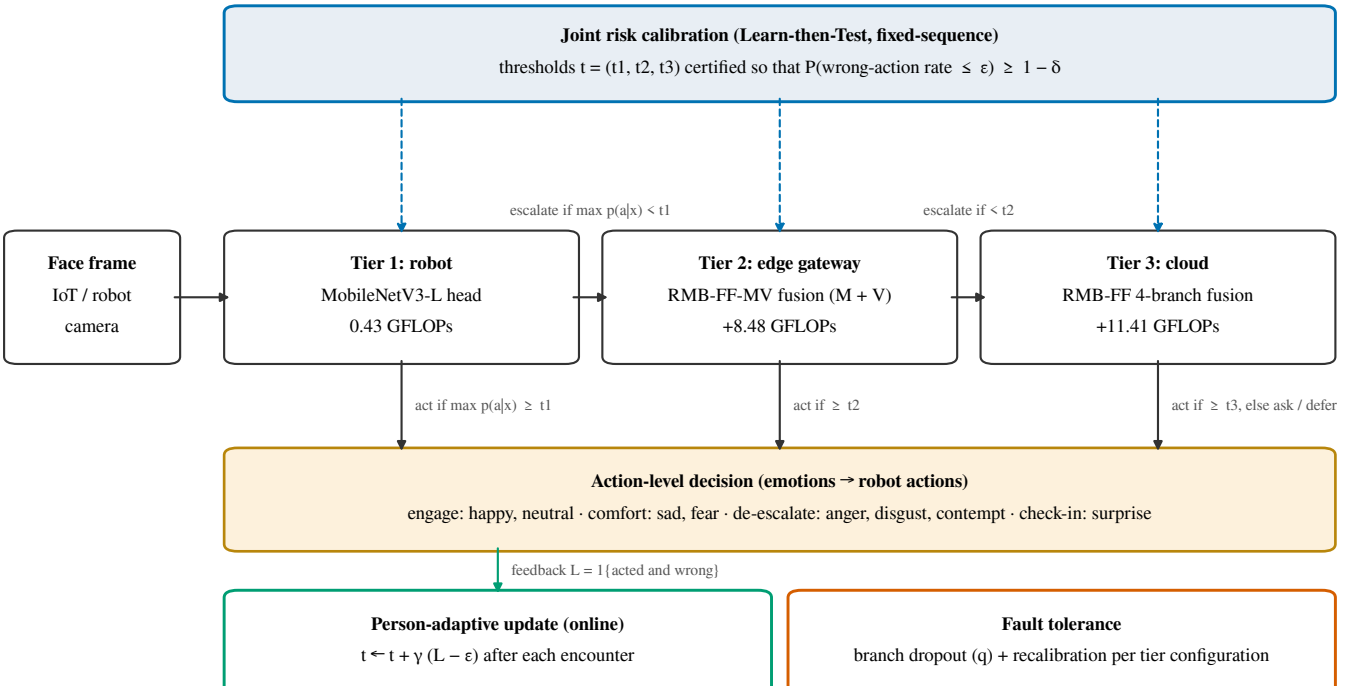
Joint risk calibration (Learn-then-Test, fixed-sequence)
thresholds t = (t1, t2, t3) certified so that P(wrong-action rate ≤ ε) ≥ 1 − δ
escalate if max p(a|x) < t1
escalate if < t2
Face frame
IoT / robot camera
Tier 1: robot
MobileNetV3-L head
0.43 GFLOPs
Tier 2: edge gateway
RMB-FF-MV fusion (M + V)
+8.48 GFLOPs
Tier 3: cloud
RMB-FF 4-branch fusion
+11.41 GFLOPs
act if max p(a|x) ≥ t1
act if ≥ t2
act if ≥ t3, else ask / defer
Action-level decision (emotions → robot actions)
engage: happy, neutral · comfort: sad, fear · de-escalate: anger, disgust, contempt · check-in: surprise
feedback L = 1{acted and wrong}
Person-adaptive update (online)
t ← t + γ (L − ε) after each encounter
Fault tolerance
branch dropout (q) + recalibration per tier configuration